Lab Experiment 1

Exploring Achievements in Engineering and Technology and Identifying Key Potentials for Success

Learning Outcomes

By the end of these activities, students will be able to:

1. Set clear personal and professional goals and develop a plan for academic and career success.
2. Research and evaluate engineering fields, career opportunities, and educational requirements, including tuition costs.
3. Compare technology and engineering programs across colleges and understand the differences between technician, technologist, and engineer roles.
4. Reflect on your engineering interests, motivations, past job experiences, and plans for further education or transfer.
5. Identify activities to explore engineering careers and assess your potential to succeed in the field.

Lab exercise

Objective:
To deepen your understanding of engineering and technology through reflection, research, and planning activities.

Instructions:
Complete the following questions from **Chapter 1:** Achievements in Engineering and Technology Study

Material prepared by prof. L. Wu Arenas

Self-reflection activities:

- Write your personal and professional goals.
- What strategies will you use to strengthen your commitment to achieving your goals.
- Determine and provide a brief explanation of **five** technical skills that you think you must have to be successful in your career.

Career research activities:

- Research **four** different jobs available for your program of study after you graduate and identify from your course flowsheet **three** classes that you consider will be important for that job, provide explanation for each class.
- Choose one of the jobs from the previous question and answer the following questions:
- Was the role description as you imagined it for that specific job?
- If you were to pursue that job after graduation, what would your plan be to develop your professional career?

College readiness activities:

- Using the following transfer explorer tool:
    - CUNY to CUNY by Course - Transfer Explorer

  Select **two** introductory courses from your program of study and search for **one** CUNY college that provides elective transfer credit for them. Provide the course code and description.

- Following up on the previous question, what differences do you notice about the course from your flowsheet and the transfer course, are there any prerequisites or grade requirements to satisfy it? Explain.

Reflection and transparency

Prepare a brief reflection that includes:

- The AI tool used and how (If you used one)
- What you learned while doing this lab

Grading Rubric

- **Objectives:** Identifies objectives and provides the necessary background or context (0-3 points)
- **Results/Analysis:** Presents results clearly. Analyzes findings correctly, with proper calculations and/or comparisons as needed. (0-6 points)
- **Discussion & Reflection:** Explains significance of results, discusses errors and connects outcomes to engineering/technology concepts. (0-5 points)
- **Professionalism & Format:** Report is well-organized, follows the provided template, uses clear language, proper grammar and formatting. Includes citation and references (if outside sources were used) (0-6 points)
- **Total:** 20 points

Deliverables

- Submit your completed activity in a lab report following the lab template.